

Project Proposal: New MEMIT Benchmarks

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Paper Details

Paper Name: Mass Editing Memory in a Transformer by Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, David Bau

- **Paper Link:** <https://arxiv.org/abs/2210.07229>
- **Codebase:** <https://github.com/kmeng01/memit>

MEMIT introduces a method for directly editing factual memories in large language models by updating transformer MLP weights across a range of critical layers, scaling to thousands of simultaneous edits on GPT-J (6B) and GPT-NeoX (20B).

To expand on MEMIT, we'll perform a baseline study and a model ablation/analysis study. MEMIT will be adapted to run on Llama via HuggingFace, and potentially other models if time allows (the best options include Mistral (7B) or Pythia). This study will use 1–2 GPUs. Based on the paper's appendix, we estimate each full run (10k edits) takes approximately 7–8 hours on a single A100, and total disk space will not exceed 50 GB.

Baselines

The paper compares MEMIT against FT-W (fine-tuning with weight decay), MEND (hypernetwork-based editing), and sequential ROME. We will reproduce these baselines on Llama and additionally vary the following hyperparameters:

- The paper edits layers $R = \{3,4,5,6,7,8\}$ for GPT-J, identified by causal tracing. We want to know whether this range is optimal and whether it generalizes to Llama and other HuggingFace models.
- The covariance hyperparameter λ is set to 15,000 for GPT-J. This controls the tradeoff between inserting new associations and preserving old ones.
- The number of edits n is tested up to 10,000 in the paper. We can test scaling behavior at different values of n .

Our baselines are as follows:

- Layer sweep:** One option is to empirically vary R from a single layer (analogous to ROME) up to the default 6-layer range and wider, both on Llama and on a second HuggingFace model, to determine whether the optimal layer count is model-dependent.
 - If time allows, for a stronger test, we could run causal testing on Llama (and another model) to mechanically identify what layers store factual associations and compare the casual tracing prediction against the empirical evidence.

- b. **λ sweep:** One option is to test λ across a range of values (ex: 1000, 5000, 15000, 50000) on Llama to see whether the paper's default transfers across architectures, and measure the tradeoff between efficacy and specificity at each value.
 - i. When testing with different models, we will need to normalize by the Frobenius norm of the edited layer so the sweep values are comparable across architectures.
 - 1. This would tell us Llama might need 40,000, but this could indicate Llama is just different from GPT-J or the weights are scaled differently. If it's the latter, we see GPT-J needs $\lambda = 15,000$ and Llama might need $\lambda = 40,000$, but after normalizing this could be the *same effective strength*.
- c. **Edit count scaling:** One option is to run MEMIT at $n = 100, 1000, 5000$, and $10,000$ edits on Llama and compare the scaling curve against the paper's GPT-J results to determine whether degradation occurs at the same rate.
 - i. To know which edits degrade, we can track regency bias (do earlier edits get overwritten) and subject collision (do edits that share subjects interfere more than unrelated edits?).

Ablation/Analysis Study

- a. **Jailbreaking:** *Will the model hallucinate the old (correct) information?* MEMIT is only evaluated on clean prompts (i.e. Micheal Jackson plays the sport of?). We want to test whether prompt injections or adversarial prompts can cause the model to revert to its original pre-edit knowledge.
 - i. Prepend a passage that asserts the original true fact, then ask the question.
 - ii. Provide several examples that demonstrate the fact, then ask the question.
 - iii. Instruct the model to take on a persona that would know the original fact.
 - iv. Prompt Injection: Assert the old fact, then ask the question.
- b. **Cross-domain testing:** *What happens when we switch domains of knowledge?* We have Batch A (what sport each athlete plays) and then Batch B (what country each athlete is from), where the subject (athlete) is shared but the relation (sport vs. country) differs. For control, we have Batch C, which has completely unrelated edits with different subjects. We test whether inserting Batch B degrades the model's retention of Batch A by reporting the domain-specific scores used in the paper (ES, PS, and NS). We test three different conditions:
 - i. Sequential Editing (A then B)
 - ii. Joint Editing (A and B simultaneously with MEMIT)

- iii. Control (C and then A)

Project Timeline

Week of Mar 24 - Get MEMIT running on Princeton's cluster and port it to Llama via HuggingFace. Verify basic editing works with a small sanity check.

Week of Mar 31 - Run causal tracing on Llama. Begin baseline (a): sweep over layer configurations at $n = 1000$ and $n = 10,000$ on COUNTERFACT.

Week of Apr 7 - Complete baseline (a) on a second HuggingFace model. Run baselines (b) and (c): λ sweep and edit count scaling on Llama.

Week of Apr 14 - Run ablation (a): probe the edited model with adversarial prompts and measure reversion rates.

Week of Apr 21 - Run ablation (b): cross-domain Theory A and Theory B experiments across all conditions.

Week of Apr 28 - Write paper: Introduction, Related Work, Baseline Results, Methods, Layer and λ Analysis, Ablation Results, Conclusion

Week of May 5 - Compile and format paper, finalize poster presentation (figure out printing if needed), submit.